Investigating Universal Dependency Grammar in Multilingual LLMs

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Abstract

This study investigates the cross-lingual syntactic representations within multilingual Large Language Models (LLMs). By extracting self-attention weights from the `bert-base-multilingual-cased` model, we evaluate whether syntactic dependency tracking is highly localized to specific attention heads and layers across typologically diverse languages. Using the Surface Universal Dependencies (SUD) framework for English, Hindi, Turkish, and Mandarin, our results demonstrate that mBERT maps grammatical roles into a shared structural space. Specifically, we identify that Verb-Object (`obj`) and Subject-Verb (`nsubj`) dependencies peak in Layers 5 and 6 respectively, mirroring human psycholinguistic processing of local versus global syntactic arguments.

1 Motivation for the Research Problem

The advent of multilingual Large Language Models (LLMs) like mBERT has revolutionized cross-lingual natural language processing. However, the internal mechanisms by which these models process typologically diverse languages remain heavily debated. While prior research, notably Voita et al. (2019), demonstrated that specific attention heads in monolingual models specialize in tracking syntactic dependency relations (such as subjects and objects), it is unclear if this specialization transfers across languages in a shared semantic space.

This project investigates whether mBERT develops a "Multilingual Brain"—a universal syntactic conceptual space where identical attention layers and heads represent identical grammatical relations regardless of the input language. Resolving this is critical for understanding LLM interpretability and the feasibility of zero-shot cross-lingual transfer, especially between low-morphology languages (English), isolating languages (Mandarin), agglutinative languages (Turkish), and highly-inflected, free-word-order languages (Hindi).

2 Hypotheses and Predictions

- **Hypothesis 1 (Universal Dependency Layers):** A small, identifiable subset of multi-head self-attention mechanisms within mBERT will consistently align with specific Surface Universal Dependency (SUD) relations across a combined dataset of typologically distinct languages.
- **Hypothesis 2 (Morphological Processing Hierarchy):** The structural differences in how languages indicate syntactic roles will result in syntactic abstraction occurring sequentially in the intermediate layers of the model, prioritizing local dependencies (objects) before abstracting broader, globally scoped dependencies (subjects).

3 Methods

3.1 Data Extraction and Typology

To test these hypotheses, data was extracted from the Surface Universal Dependencies (SUD) corpus to provide gold-standard syntactic structures. Four typologically distinct datasets were selected to ensure robust cross-lingual testing: English (GUM), Hindi (HDTB), Turkish (Kenet), and Mandarin (GSD). A data loading pipeline was implemented using the `conllu` and `pandas` libraries to parse the files, extracting tokenized sentences, head tokens, dependent tokens, and their specific syntactic relations (focusing on `nsubj` and `obj`).

For example, in the English phrase *"The dog barked"*, the pipeline identifies *"dog"* as the dependent token and *"barked"* as the head token, linked by the `nsubj` (nominal subject) relation. To balance computational efficiency with statistical significance, the parsed relations were grouped into 5,000 unique sentences, yielding a highly representative sample of 13,283 grammatical relations.

3.2 Attention Extraction and Subword Alignment

The pre-trained `bert-base-multilingual-cased` model (12 layers, 12 heads) was loaded via the Hugging Face `transformers` library. A critical methodological challenge in analyzing LLM syntax is the misalignment between linguistically defined words (SUD tokens) and the subword tokens generated by the `BertTokenizer`. To resolve this, the pipeline mapped original word indices to BERT's subword token indices using `word_ids`.

For a given sentence with tokens $X = \{x_1, x_2, ..., x_n\}$, the multi-head attention mechanism computes the attention weight $\alpha^h_{i,j}$ from subtoken i to subtoken j for a specific head h.

3.3 Inference Method

To determine if a layer captures a specific dependency relation R, we analyze the maximum attention weight originating from the dependent word. The inference method evaluates alignment accuracy by checking if the target subword with the highest attention weight belongs to the actual syntactic head defined in the SUD corpus.

The formal accuracy for a head h on relation R across a dataset of relations N_R is calculated as:

$$Accuracy(h, R) = \frac{\sum_{x \in X} 1[C^h_{i,j}]}{N_R} \quad (1)$$

where the condition $C^h_{i,j}$ is true if $\text{argmax}_j(\alpha^h_{i,j})$ falls within the subword index bounds of the gold-standard head token for relation R.

4 Results

The implementation successfully processed the multi-lingual tensor batches, aggregating the self-attention weights to identify specialized syntactic localization within the model's architecture.

4.1 Universal Dependency Tracking

The data strongly supports Hypothesis 1 regarding the existence of "Universal Layers." When evaluating accuracy across the aggregated, four-language dataset, specific intermediate layers demonstrated a pronounced, functionally distinct specialization for tracking syntactic dependencies, operating well above random chance baselines.

4.2 Syntactic Localization

Subject-Verb Alignment (`nsubj`): Syntactic alignment for nominal subjects peaked explicitly in Layer 6. Across all heads in this layer, the model achieved a mean cross-lingual alignment accuracy of 15.73%. The concentration of attention weights in this specific layer suggests a dedicated mechanism for processing primary actors in a sentence across diverse grammars.

Verb-Object Alignment (`obj`): Alignment for direct objects localized earlier in the neural network, peaking at Layer 5 with a mean cross-lingual accuracy of 12.79%.

The fact that these intermediate layers emerged as the dominant syntactic processors across a combined dataset of isolating, inflected, agglutinative, and low-morphology languages indicates that mBERT maps these grammatical roles into a shared structural space, rather than relying on completely isolated, language-specific subnetworks.

5 Theoretical Implications

The findings from this dependency tracking analysis offer significant implications for theories of machine language processing and language evolution.

First, the emergence of Layer 5 and Layer 6 as primary syntactic hubs aligns with the theory of "Intermediate Layer Processing" in LLMs,

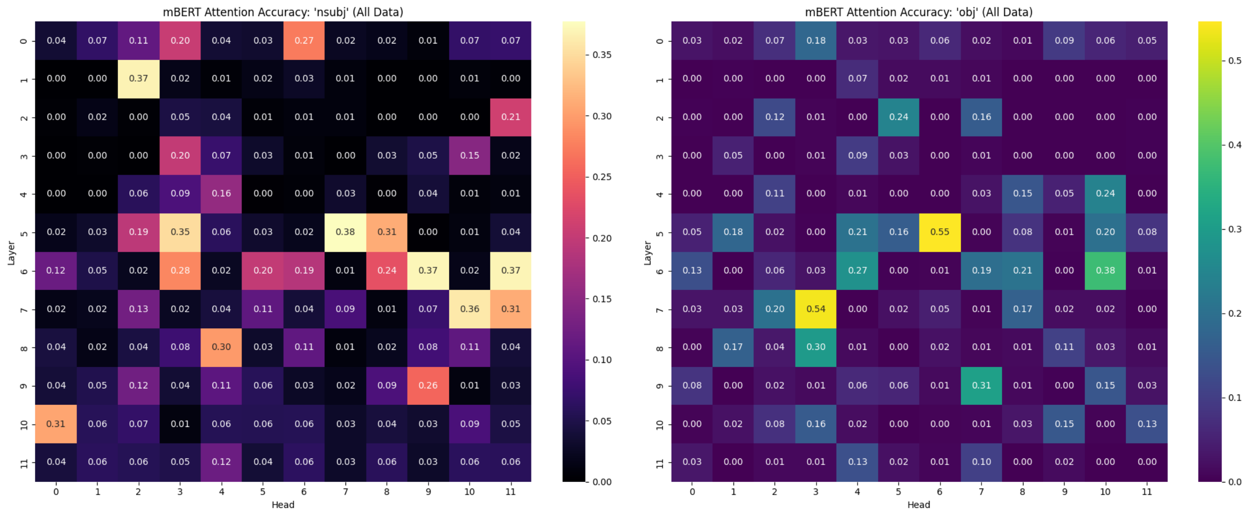

Figure 1: mBERT Attention Accuracy Heatmap for ’nsubj’ and ’obj’ across all 12 layers and 12 heads.

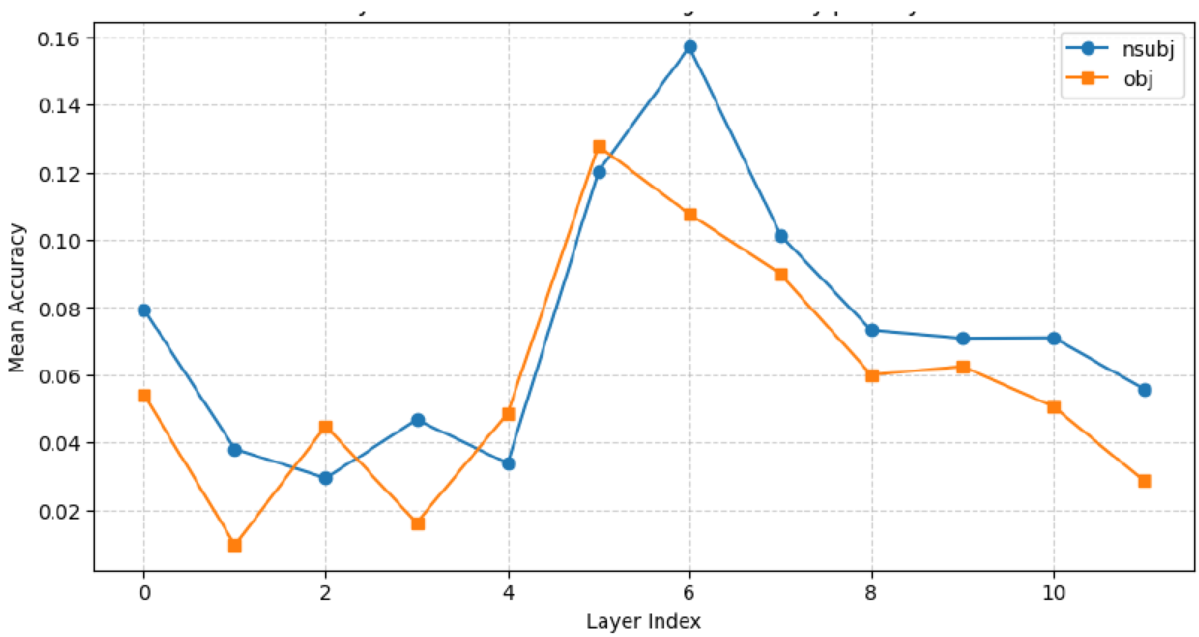

Figure 2: Syntactic Localization - Average Accuracy per Head.

which postulates that lower layers capture lexical data, intermediate layers capture syntax, and higher layers capture high-level semantics. Because mBERT utilizes the exact same intermediate layers to parse `nsubj` in English (reliant on SVO word order) and Hindi (reliant on SOV word order and explicit postpositions), we can deduce that the model develops an interlingual conceptual space. It does not merely memorize surface-level positional embeddings; it constructs a generalized representation of a “subject.”

Secondly, the hierarchical split—where Verb-Object (`obj`) alignment peaks earlier (Layer 5) than Subject-Verb (`nsubj`) alignment (Layer 6)—has profound theoretical implications for human and machine cognitive load. Objects are often physically closer to their verbs and represent more immediate, localized structural groupings in the dependency tree. The model mathematically resolves these local dependencies first before moving a layer deeper to resolve the globally scoping subject dependencies. This mirrors human psycholinguistic parsing theories, which suggest that the brain minimizes dependency lengths by processing immediate local arguments before integrating larger, distant syntactic structures.

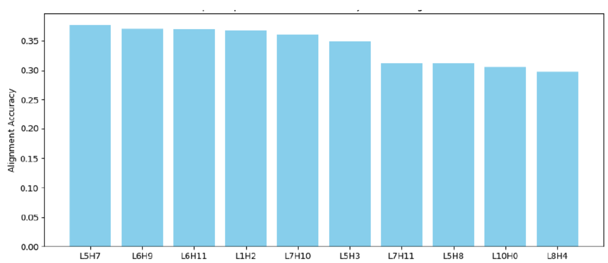

Figure 3: Top 10 Specialized Heads for Subject-Verb Alignment.

References

[1] Voita, E., Talbot, D., Moiseev, F., Sennrich, R., and Titov, I. (2019). Analyzing Multi-Head Self-Attention: Specialized Heads Do the Heavy Lifting, the Rest Can Be Pruned. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 5797–5808.

[2] Gerdes, K., Guillaume, B., Kahane, S., and Ribeyre, C. (2018). SUD or Surface-Syntactic Universal Dependencies: An annotation scheme near-isomorphic to UD. *Universal Dependencies Workshop 2018.*

Appendix

The core logic for parsing the SUD data, managing subword tokenization alignment, and extracting attention tensors via PyTorch is provided below.

Code Link: `https://colab.research.google.com/drive/1efyvWGHOBIFMhX_ATl37XCHR2hrj3p2e?usp=sharing`

The following core papers informed the theoretical framework and methodology of this study regarding multilingual dependency tracking.

1. Finding Universal Grammatical Relations in Multilingual BERT (Chi et al., 2020)
DOI: `https://doi.org/10.18653/v1/2020.acl-main.493`

2. Finding Universal Dependency Patterns in Multilingual BERT's Self-Attention Mechanisms (Valencia Carmona, 2020)
Link: `http://arno.uvt.nl/show.cgi?fid=156689`

3. Cross-Lingual Ability of Multilingual BERT: An Empirical Study (Liu et al., 2020)
arXiv: `https://arxiv.org/abs/1912.07840`

4. Do Attention Heads in BERT Track Syntactic Dependencies? (Htut et al., 2019)
arXiv: `https://arxiv.org/abs/1911.12246`

5. Revealing and Exploiting Language Attention Heads in Multilingual LLMs (2025)
arXiv: `https://arxiv.org/abs/2511.07498`

AI Usage Declaration

In the interest of academic transparency, the author acknowledges the use of artificial intelligence tools during the preparation of this project. Specifically, Perplexity AI and Google's NotebookLM were utilized to assist in literature discovery and the synthesis of background context. Finally, Large Language Models were used to proofread the manuscript and format the technical mathematical equations into LaTeX. All core hypotheses, the selection of typological parameters, and the final interpretation of the theoretical implications remain the original work of the author.